

Assignment 2

Into To SE

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Repository: <https://github.com/ittzzzz-ahmed/legacy-calc-2009.git>

1. Introduction

This assignment required refactoring a legacy C++ loan calculator project, organizing it using modern software engineering practices, adding documentation, unit tests, and maintaining a GitHub repository with proper commits

The original project contained:

- Mixed UI and business logic
- Hardcoded values
- No testing
- No documentation
- Outdated Qt dependencies
- Poor structure

2. Repository Setup

Steps followed:

1. Forked the starter repository on GitHub and named it legacy-calc-2009.
2. Added a .gitignore file to exclude build artifacts
3. Created a dev branch to commit development work.
4. Cloned the fork locally and configured Git.

3. Code Refactoring

The refactoring focused fixing bugs, improving safety, and reorganizing structure.

3.1 New Project Structure

```
Muhammed Ahmad@Ahmed-PC MINGW64 ~/Desktop/legacy-calc-2009 (dev)
$ ls
Doxyfile.unknown  README  config.txt  src/  test/
```

```
Muhammed Ahmad@Ahmed-PC MINGW64 ~/Desktop/legacy-calc-2009/src (dev)
$ ls
Loan.cpp  Loan.h  main.cpp
```

3.2 Fixes and Improvements

Removed Qt dependency

Legacy UI files LoanCalcQtMainWindow.cpp, .h, .pro were removed.

Replaced hardcoded values

Introduced **config.txt**:

default_amount=100000

default_interest=10

default_years=5

Introduced input validation

Invalid cases (negative amount, zero years, negative interest) now return false.

Safer data types

Used long double to avoid overflow in EMI calculation.

Bug Fixes

- Fixed overflow risk in pow()
- Fixed EMI formula
- Ensured correct output formatting

4. Unit Testing (GoogleTest)

Unit tests were added to verify EMI calculations and input validation.

Tests Included

1. NormalEmiCalculation – verifies EMI formula
2. InvalidInputHandling – rejects invalid input
3. LargeTenureNoOverflow – ensures no overflow on long durations

Test Output

PASSED 3 tests

```

M /c/Users/LENOVO/legacy-calc-2009/src
LENOVO@mubashir MINGW64 ~
$ cd /c/Users/LENOVO/legacy-calc-2009/src
LENOVO@mubashir MINGW64 /c/Users/LENOVO/legacy-calc-2009/src
$ ls
Doxyfile  Loan.cpp  Loan.h  Loan.h.gch  LoanTest.cpp  docs  googletest  main.cpp  main.exe
LENOVO@mubashir MINGW64 /c/Users/LENOVO/legacy-calc-2009/src
$ g++ -std=c++17 Loan.cpp LoanTest.cpp -lgtest -lgtest_main -pthread -o loan_tests.exe
LENOVO@mubashir MINGW64 /c/Users/LENOVO/legacy-calc-2009/src
$ ./loan_tests.exe
[-----] Running 3 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 3 tests from LoanTest
[ RUN    ] LoanTest.NormalEmiCalculation
[ OK     ] LoanTest.NormalEmiCalculation (0 ms)
[ RUN    ] LoanTest.InvalidInputHandling
[ OK     ] LoanTest.InvalidInputHandling (0 ms)
[ RUN    ] LoanTest.LargeTenureNoOverflow
[ OK     ] LoanTest.LargeTenureNoOverflow (0 ms)
[-----] 3 tests from LoanTest (12 ms total)

[-----] Global test environment tear-down
[-----] 3 tests from 1 test suite ran. (27 ms total)
[ PASSED ] 3 tests.
LENOVO@mubashir MINGW64 /c/Users/LENOVO/legacy-calc-2009/src
$ |
```